

Ex: Lays chips $\longrightarrow$ ₹ 20
 $\downarrow$
 50 gms (PM) $\textcircled{\geq \uparrow}$

NH: The avg weight of chips packets = 50 gms
 (PM = SM)

AM: The avg weight of chips packets < 50 gms
 (SM < PM) ✓

Ex: Peanut Butter $\longrightarrow$ may 2 gms of fat
 per 100 gms of
 butter

NH: The avg fat per 100 gms $\leq$ 2 gms (PM $\geq$ SM)

AM: The avg fat per 100 gms > 2 gms.
 (SM > PM)

500 packets

Ex1:

PM
 (50 gms)

① 50.001

② 50

③ 49.998

⋮

⑤00 50.03

SM = 49 gms

PM

vs

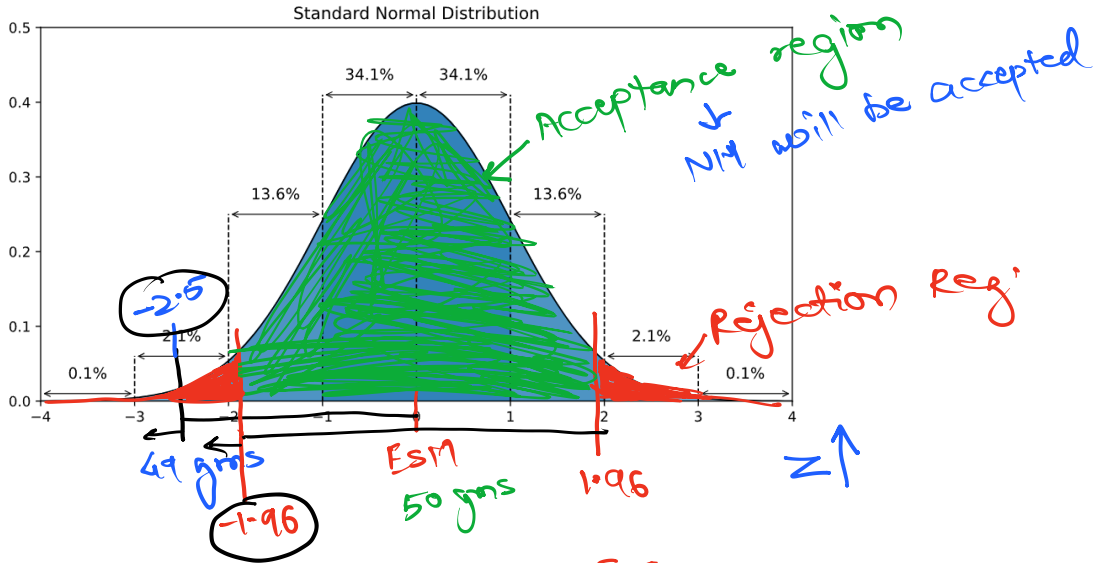
SM

Z-test:

CL
CI

95% Conf

-ve
+ve



PM (50 gms) vs SM (49 gms)

49.5 gms

49.9 gms



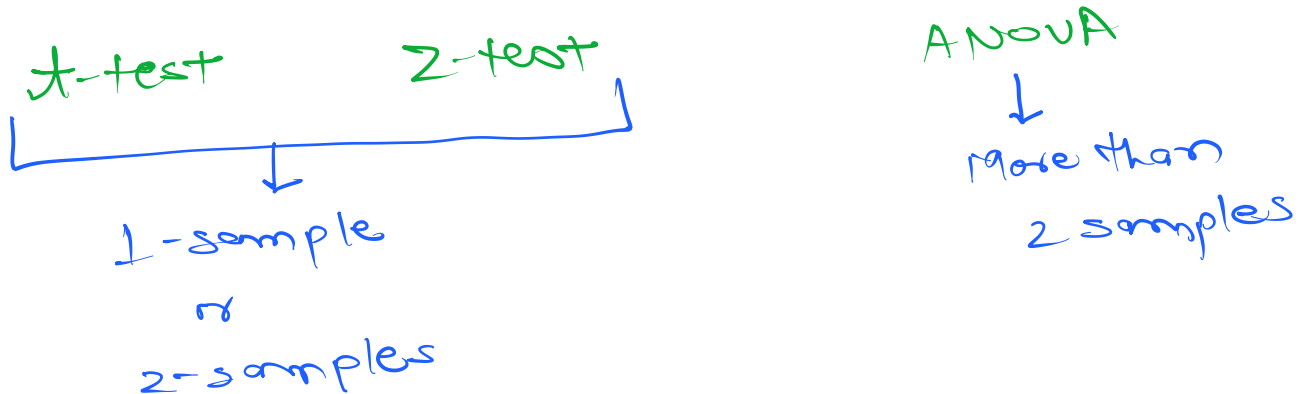
Test statistic comes into picture to help us decide whether the diff is significant or not.

$$\begin{aligned} \underline{\text{Z-stat}} &= \frac{\text{SM} - \text{PM}}{\text{PSD} / \sqrt{SS}} \\ &= \frac{49 - 50}{\text{PSD} / \sqrt{100}} \\ &= \underline{\underline{-2.5}} \end{aligned}$$

Test-statistic is used to quantify the diff

blw what we expected vs what we got.

Numerical sample



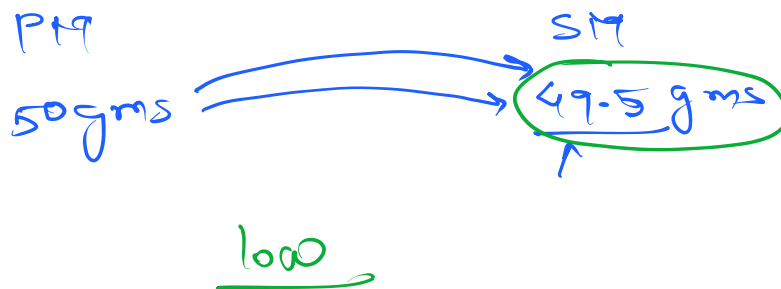
Categorical sample

↓
Chi-squared test

Definition: The probability of observing data as extreme as your results, assuming the null hypothesis is true.

The average weight of the bars is equal to the target. (The machine is working correctly).

↓



$$p\text{-value} = 0.0062 \times 100 \\ = \underline{0.62\%}$$

$$\downarrow \\ \underline{99.38\%}$$

There is only 0.62% prob that we get sm of 49.5 gms if Null Hyp was true

→ if 0.62% p-value < ^{5%} significance level (α)

$\downarrow$
Reject the H_0

$$\downarrow \\ \underline{99.38\%}$$

$$\begin{aligned} \text{Significance level} &= 1 - \text{confidence level} \\ &= 1 - 0.95 \\ &= \underline{0.05 (5\%)} \end{aligned}$$